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From: Siena Romano, Pennsylvania State University MAS Student Consultant

To: Mayor Daniel Laurie and the San Francisco Traffic Stop Analysis Committee

**San Francisco Moving Violation Trends, Forecasts, and Associations with Driver Demographics**

EXECUTIVE SUMMARY

Due to San Franciso Mayor Daniel Laurie's recent Street Safety Initiative that will increase traffic stops and citations, the San Francisco Traffic Stop Analysis Committee plans to make comparisons of new monthly stop counts and driver demographics to historical traffic data. Statistical tests and models over SF moving violation data from 10 to 20 years ago generated the following conclusions for Laurie's research questions concerning possible monthly seasonality in traffic stop occurrences, the expected number of moving violations per month, and associations between receiving a traffic citation and a driver's race and gender:

As monthly stop counts between January 2007 and June 2016 were overly dispersed, a negative binomial regression (NBR) with months as the predictor and moving violations as the response was built to uncover monthly trends. The results show significant negative trends in count for June, November, and December across the ten years with 90% confidence.

An autoregressive integrated moving average (ARIMA) model using non-season AR(2) and MA(1) terms, a seasonal AR(2) term, and a seasonal period of 12 months was applied to moving violation data from July 2011 to June 2016. Using the model's predictions, standard errors, and 90% confidence, the forecast for moving violations in July 2016, a month outside of the period, would be between 3,888 and 5,416.

Three-way chi-square tests for independence and Cramer's V statistics concluded that there was a significant, moderately sized relationship between race and receiving a citation from a moving violation for male drivers between January 2007 and June 2016. Specifically, individual pairwise comparison tests and Cramer's V statistics evidenced that Asian/Pacific Islander and Black male drivers and Black and other race male drivers had significantly different associations with receiving a moving violation citation on a moderate level.

## 1.0 – PROJECT DESCRIPTION

The San Francisco Traffic Stop Analysis Committee is conducting ongoing testing over counts, trends, and recipients of moving violations due to SF Mayor Daniel Laurie's new Street Safety Initiative. The Initiative was enacted in December 2025 to increase street safety for all SF residents by coordinating city police departments and stop policies, implementing automated speed cameras, and improving streets and walkways[1]. As one of the directive's goals is to reduce traffic collisions, Laurie has increased police presence and cameras throughout the city, and he expects the number of traffic citations to rise in 2026.

Laurie believes that an increase in traffic stops can lead to disproportionate profiling and citations that are not distributed for public safety but to meet department demands. Therefore, he requested that the Committee, with the assistance of a Penn State MAS student, analyze SF traffic stop data from 10 to 20 years ago to confirm if stop disparity and spikes in stops are worse, the same, or better following the new initiative. Further, Laurie asked that a model be built on historical data to see how well it can predict an interval of the expected number of moving violations per month as he'll want to check if future stop counts are outside of the appropriate range. As the historical analyses being conducted were for exploratory research with low-risk policy impacts and cost of false positives, extreme statistical precision was not required, and 90% confidence was accepted for all statistical tests.

The study used January 2007 through June 2016 SF stop data provided by a team of researchers for the Stanford Open Policing Project who collected and standardized law enforcement stops across the United States for public large scale analyses[2]. While the full dataset included all traffic stops for moving and non-moving violations, only moving violations were analyzed as auto mechanical issues or traffic collisions are unrelated to increased policing. After the initial filtering, 563,315 records remained for testing if there was a trend in when stops occurred, if the count of moving violations could be predicted, and if there were relationships between driver race and gender and receiving a citation following a moving violation.

Laurie plans to begin comparative analyses six months after the start of the directive (May 2026) when enough recent traffic citation data is collected. As a result, the historical trend analysis, model building, and association testing must conclude in April 2026.

## 1.1 – RESEARCH QUESTIONS

1. Was there a monthly trend in counts of traffic stops for moving violations in San Francisco between 2007 and 2016?
2. Can the expected number of moving violations in San Francisco be forecasted with a prediction interval for a month outside of the data's time period, July 2016?
3. Were there three-way interactions between drivers' race and sex and receiving a traffic citation following a moving violation in San Francisco between 2007 and 2016?

## 1.2 – VARIABLES

The traffic stop data was sourced from the Stanford Open Policing Project[3] (OpenPoliceData · Streamlit) and included dates, locations, driver demographics, reason for stops, and outcomes of traffic stops that occurred in San Francisco between January 2007 and June 2016. After filtering the data for only moving violations, stop counts per month were calculated by aggregating the number of rows by year and month and log transformed to improve data normality. See Table 1.1 for the variables used in the analyses along with their descriptions, data types, and value ranges.

*Table 1.1 Traffic Stop Variables, Descriptions, Types, and Range of Values*

| Variable | Description | Type | Range/Notes |
|---|---|---|---|
| Month Year | Month and year of stop | Quantitative – Discrete; Explanatory | Jan 2007 – Jun 2016 |
| Stop Count | Number of stops | Quantitative – Discrete; Response | Natural Log; 8.122 – 8.865 |
| Subject Race | Race of driver pulled over | Categorical – Nominal; Explanatory/Control | asian/pacific islander, black, hispanic, other, white |
| Subject Sex | Sex of driver pulled over | Categorical – Binary; Explanatory/Control | male, female |
| Citation Issued | Flag indicating if the driver was cited | Categorical – Binary; Response | TRUE, FALSE |
| Reason for Stop | Reason the driver was pulled over | Categorical – Nominal; Control | Value includes "moving violation" |

## 2.0 – EXPLORATORY DATA ANALYSIS

The underlying data was explored prior to performing statistical analyses and modeling. Specifically, monthly stop counts were log transformed to reduce variance and summarized from January 2007 to June 2016 for research question 1 and from July 2011 to June 2016 for research question 2. Unlike research question 1 requesting evidence of monthly trends going back the full ten years, only five years of data were input into the time series models for research question 2 to improve forecasting accuracy with recent years and reduce the impact of outdated monthly patterns. Less years also prevented the models from overfitting to irrelevant noise.

Additionally, logged count outliers that were more than three standard deviations away from the mean were identified and removed (Sep 2013, 8.05 and Dec 2013, 8.00). Refer to Table 2.1 and Table 2.2 below for the min, max, mean, interquartile range, and standard deviation of the moving violation counts for the two time periods.

*Table 2.1 Summary Statistics of Monthly Stop Counts January 2007 – June 2016*

| Scale | Min | 1st Quartile | Median | Mean | 3rd Quartile | Max | Std Dev. |
|---|---|---|---|---|---|---|---|
| Untransformed | 3,367 | 4,507 | 4,996 | 4,912 | 5,503 | 7,081 | 786 |
| Natural Log | 8.122 | 8.413 | 8.516 | 8.499 | 8.613 | 8.865 | 0.161 |

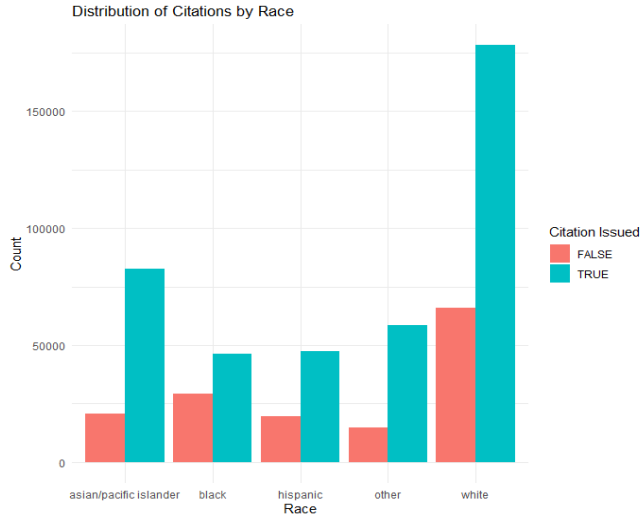
*Table 2.2 Summary Statistics of Monthly Stop Counts July 2011 – June 2016*

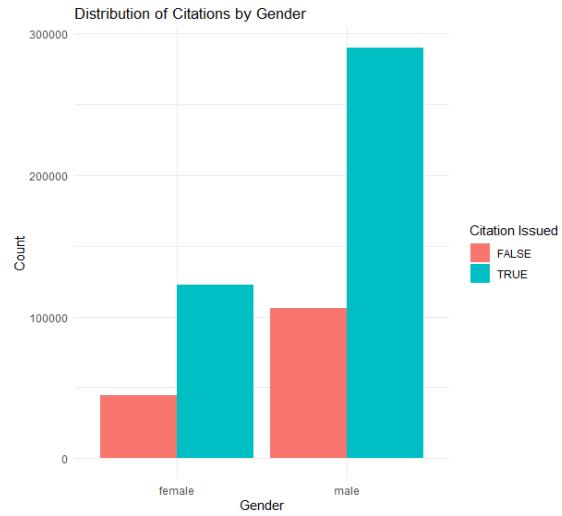
| Scale | Min | 1st Quartile | Median | Mean | 3rd Quartile | Max | Std Dev. |
|---|---|---|---|---|---|---|---|
| Untransformed | 2,989 | 3,949 | 4,500 | 4,445 | 4,951 | 6,015 | 669 |
| Natural Log | 8.003 | 8.281 | 8.412 | 8.388 | 8.507 | 8.702 | 0.154 |

Monthly traffic stop data was further evaluated to determine normality using histograms and boxplots. Refer to Figures 2.1 and 2.2 in the Appendix depicting a normal bell curve in the histograms of moving violations for the two time periods. Figures 2.3 and 2.4 in the Appendix also provide a visual representation of the range of stop counts and lack of outliers for the two time periods prior to statistical analyses. Last, the scatterplot in Figure 2.5 in the Appendix shows no visual trends that would impact modeling of monthly stops between July 2011 and June 2016.

To understand the breakdown of stop citations by driver demographic for research question 3, counts of drivers pulled over for moving violations were aggregated by gender and race and further grouped by whether a citation was given or not. Refer to Tables 2.3 and 2.4 in the Appendix for the citation counts by driver race and sex. Additionally, for a visual representation of the distribution of citations, Figures 2.6 and 2.7 below show comparative bar graphs by race and gender. Based on the distribution counts and graphs, the number of white and male drivers pulled over for moving violations were noticeably higher than the stop counts of all other drivers.

*Figures 2.6 and 2.7 Comparative Bar Graphs of Citations by Race and Gender*




## 3.0 – STATISTICAL ANALYSIS

### 3.1 – *Was there a monthly trend in counts of traffic stops for moving violations in San Francisco between 2007 and 2016?*

NBR with months as the predictor was performed to confirm if there were significant count differences in moving violations between months, regardless of the year, from January 2007 to June 2016. To test that the data passed NBR assumptions, no significant outliers were found using cook's distance as distances were under 0.5 (shown in Figure 3.1 in the Appendix). Also, a likelihood ratio test between the NBR model and a Poisson model with months as a predictor[4] confirmed that traffic stop counts were appropriately over dispersed and NBR was a better fit (chi-squared p-value of 0). Last, a heteroskedasticity-consistent covariance matrix estimator was added to the NBR model to handle autocorrelation and produce robust standard errors[5].

The NBR results shown in Table 3.1 indicate that months June, November, and December were significantly lower over the ten years. Specifically, at 90% confidence, the expected number of stops was about 15% less in June, 12% less in November, and 17% less in December in reference to the incidence rate ratio (IRR). None of the months had significantly higher stop trends. Refer to Table 3.2 in the Appendix for the NBR results for all months besides the reference group (Jan).

*Table 3.1 Significant Negative Binomial Regression Results*

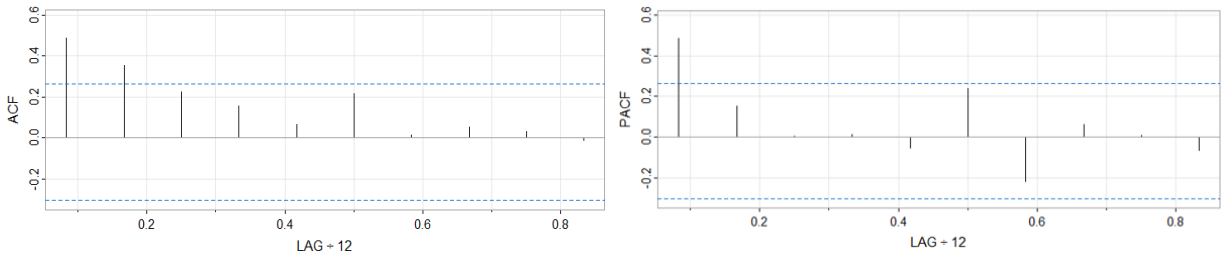
| | Jun | Nov | Dec |
|---|---|---|---|
| **Estimate** | -0.161 | -0.128 | -0.185 |
| **IRR** | 0.851 | 0.880 | 0.831 |
| **Robust Std. Error** | 0.065 | 0.077 | 0.079 |
| **P-Value** | 0.013 | 0.095 | 0.019 |

To confirm that the trends identified were truly significant and not caused by overfitting, a likelihood ratio test was conducted between the NBR model with months as a predictor and an NBR null model (with intercept only)[4]. The results showed that the model with months as a predictor was significantly different than the null model with a chi-square p-value of 0.072. Therefore, at 90% confidence, the NBR results above were significant trends.

3.2 – *Can the expected number of moving violations in San Francisco be forecasted with a prediction interval for a month outside of the data's time period, July 2016?*

An ARIMA model with SF traffic stop data from July 2011 to June 2016 was used to forecast the interval of expected number of moving violations in July 2016. Specifically, an ARIMA model with non-season AR(2) and MA(1) terms, a seasonal AR(2) term, and a seasonal period of 12 had the lowest MAE (0.094), MAPE (0.011), and RMSE (0.125) in comparison to other ARIMA, GARCH, and triple exponential smoothing models. Additionally, the selected model's results passed the assumptions of no trend in the scatterplot, no significant autocorrelations in the ACF of residuals, normally distributed residuals in the QQ plot, and non-significant residuals in the Ljung-Box plot[6] shown and further discussed in Figure 3.2 in the Appendix. These plots, along with the ACF and PACF shown in Figure 3.3 depicting stationary data (white noise) with lags tailing to zero[6], confirm that the ARIMA model was appropriate for the data.

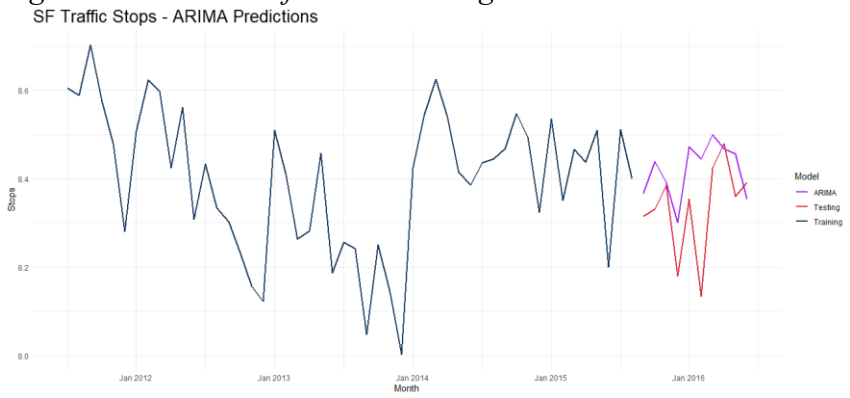
*Figure 3.3 ACF and PACF*




Before forecasting the prediction interval for July 2016, the ARIMA model was trained on moving violation data from July 2011 to August 2015 (85% of the stops). Figure 3.4 shows a line

plot of the ARIMA model's average forecast for each stop count compared to the true test set[7] (stop counts from September 2015 to June 2016).

*Figure 3.4 Line Plot of ARIMA Average Forecasts in the Test Set*




The ARIMA model forecasted 12 months of moving violations using the July 2011 to June 2016 data. Figure 3.5 in the Appendix shows those prediction intervals with one to two standard deviations from July 2016 to June 2017. After applying the exponential function to the natural log counts, the upper and lower bounds for July 2016 were calculated using a predicted value of 4,652 and a margin of error of 764 and confirmed at 90% confidence that the count of moving violations would be between 3,888 and 5,416.

3.3 – *Were there three-way interactions between drivers' race and sex and receiving a traffic citation following a moving violation in San Francisco between 2007 and 2016?*

To measure the association between driver sex, race, and receiving a traffic citation following a moving violation between January 2007 and June 2016, three-way chi-square tests for independence and Cramer's V statistics were calculated to determine if the variables were conditionally associated and not the same across driver demographics and to conclude the strength of those interactions. These tests were appropriate as the data passed key requirements of expected counts greater than five and observations were collected independently[8] with rows representing unique drivers pulled from all areas of the city by different police officers. The chi-square tests resulted in p-values of approximately 0 for all relationships, meaning there were significant differences in the associations between driver race and receiving a citation across male and female drivers and between driver gender and receiving a citation across all races. The Cramer's V statistics were also analyzed because it determined the true effect size of the

interactions and is necessary when performing chi-square tests that are highly sensitive to large sample sizes (563,315).

The results in Table 3.3 show that effect size of the associations between driver gender and receiving a citation were very low across all races given a degree of freedom of 1 and results below 0.1, the low effect size threshold. This means that the relationship strengths were weak[9] and the associations between driver gender and receiving a citation across all races were likely independent and not changing across race levels.

*Table 3.3 Association of Driver Gender and Receiving a Citation Given Race*

| | **Asian/Pacific Islander** | **Black** | **Hispanic** | **Other** | **White** |
|---|---|---|---|---|---|
| **Cramer's V** | 0.0159 | 0.0339 | 0.0219 | 0.0170 | 0.0084 |

Further, the results in Table 3.4 show that the effect size of the associations between driver race and receiving a citation for males were medium with degrees of freedom of 4 and results around 0.15, the medium effect size threshold. This means that the magnitudes of the differences in associations for males were moderate[9], indicating that the associations between driver race and receiving a citation can change for male drivers. The effect size was low for females as the statistic was further under 0.15.

*Table 3.4 Association of Driver Race and Receiving a Citation Given Gender*

| | **Female** | **Male** |
|---|---|---|
| **Cramer's V** | 0.1123 | 0.1415 |

After identifying that there was a medium sized effect in the relationship between race and receiving a citation for males, pairwise comparisons between races and citations were performed to determine which associations between race and receiving a citation were significantly different. Based on the results, all races had significant differences in the associations between each other and receiving a citation with p-values of approximately 0. Using the Cramer's V statistics to measure the strength of those associative differences, Table 3.5 in the Appendix shows that the relationship between Asian/Pacific Islander and Black male drivers and receiving a citation and the relationship between Black and other raced male drivers and receiving a citation had approximate medium effect sizes (statistics around 0.3). This means that the magnitudes of the differences in associations were moderate[9] and the associations between Asian/Pacific Islander and Black drivers and between Black and other raced drivers and receiving a citation were different for male drivers during the time period.

## 4.0 – RECOMMENDATIONS

1. *Was there a monthly trend in counts of traffic stops for moving violations in San Francisco between 2007 and 2016?*

Based on the NBR results, the count of moving violations in June, November, and December were significantly lower than other months between January 2007 and June 2016. Specifically, June had about 15% less, November had about 12% less, and December had about 17% less stops, indicating negative trends for those months.

2. *Can the expected number of moving violations in San Francisco be forecasted with a prediction interval for a month outside of the data's time period, July 2016?*

Using monthly stop data between July 2011 to June 2016, an ARIMA model predicted that July 2016 would have between 3,888 and 5,416 moving violations with 90% confidence.

3. *Were there three-way interactions between drivers' race and sex and receiving a traffic citation following a moving violation in San Francisco between 2007 and 2016?*

Chi-squared tests for independence and Cramer's V statistics provided evidence that there was a moderately strong and significant three-way relationship between race and receiving a citation for male drivers. Further analyses using pairwise comparisons and Cramer's V statistics confirmed that there were significant differences in the associations between Asian/Pacific Islander and Black male drivers and receiving a citation and between Black and other raced male drivers and receiving a citation on a moderate level.

## 5.0 – RESOURCES

While completing statistical testing, modeling, and output interpretations for the report's research questions, the following academic books, papers, and R and SAS tutorials were referenced:

| | |
|---|---|
| Negative Binomial Regression | • *Regression Analysis of Count Data* by A. Colin Cameron and Pravin K. Trivedi<br>• "Negative Binomial Regression \| R Data Analysis Examples" from UCLA's |
| ARIMA Model Forecasting | • *Forecasting: Principles and Practice* by Rob J Hyndman and George Athanasopoulos<br>• "Introducing ARIMA Models" by Joshua Noble |
| Chi-Square Three-Way Test of Association and Cramer's V Statistic | • *Statistical Notes for Clinical Researchers: Chi-Squared Test and Fisher's Exact Test* by Hae-Young Kim<br>• *An Introduction to Categorical Analysis* by Alan Agresti |

## 6.0 – CONSIDERATIONS

### 6.1 – POLICY AND OPERATIONAL CONSIDERATIONS

SF law enforcement policy changes between January 2007 and June 2016 related to department quotas, traffic stops, or citations could be a lurking variable that impacted trend analysis or independence testing across the ten years. For example, there could have been a significant trend in when tickets were given out some months and not all because of changes in department objectives. Further, changes in policy around ticketing residents of different races and sexes could have impacted independence testing of these features and receiving a citation.

Additionally, the ARIMA model was not trained on all ten years of data due to the high variance in stop count across the ten years. A better predictive time series model could be produced with more data points if Laurie and the SF Traffic Stop Analysis Committee removed years with policies aimed at reducing traffic stops. This is because the years with initiatives to reduce stops do not represent the trend of increased traffic citations that are expected with the new directive.

### 6.2 – METHODOLOGICAL CONSIDERATIONS

Future analysts should note that there is an imbalance in counts between driver race and sex. While the Cramer's V statistic measures the effect size independently of total sample size, the test can be influenced by smaller or larger counts in groups and indicate a higher association. However, as the results did not show a very high effect size between any of the groups, there is no large concern that historical findings were inaccurate.

Further, while significant differences were identified in the interactions between Asian/Pacific Islander and Black male drivers and receiving a citation and between Black and other raced male drivers and receiving a citation on a medium sized level, the results do not tell us which race increases or decreases the chances of receiving a ticket. Specifically, the current study only presents the differences and magnitudes of the relationship between races and receiving a moving violation. Future research would be required to identify which races were most at risk of receiving a moving violation between January 2007 and June 2016.

## 7.0 – APPENDIX

## 7.1 – TABLES

*Table 2.3 Observation Count of Driver Race When a Citation was Issued or Not*

| | Asian/Pacific Islander | Black | Hispanic | Other | White |
|---|---|---|---|---|---|
| **Cited** | 82,557 | 46,130 | 47,427 | 58,416 | 178,236 |
| **Not Cited** | 20,823 | 29,352 | 19,625 | 14,716 | 66,033 |
| **Total** | 103,380 | 75,482 | 67,052 | 73,132 | 244,269 |

*Table 2.4 Observation Count of Driver Gender When a Citation was Issued or Not*

| | Male | Female |
|---|---|---|
| **Cited** | 290,155 | 122,611 |
| **Not Cited** | 106,009 | 44,540 |
| **Total** | 396,164 | 167,151 |

*Table 3.2 Negative Binomial Regression Results*

| | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Estimate** | -0.053 | 0.022 | -0.056 | 0.008 | -0.161 | -0.082 | -0.058 | -0.028 | -0.085 | -0.128 | -0.185 |
| **IRR** | 0.948 | 1.022 | 0.946 | 1.008 | 0.851 | 0.921 | 0.944 | 0.972 | 0.918 | 0.880 | 0.831 |
| **Robust Std. Error** | 0.069 | 0.068 | 0.060 | 0.070 | 0.065 | 0.059 | 0.076 | 0.080 | 0.073 | 0.077 | 0.079 |
| **P-Value** | 0.440 | 0.745 | 0.352 | 0.915 | 0.013 | 0.163 | 0.448 | 0.723 | 0.241 | 0.095 | 0.019 |

*Table 3.5 Pairwise Comparison of Race and Receiving a Citation with Cramer's V Statistics*

| Paired Races | Cramer's V |
|---|---|
| Asian/Pacific Islander; Black | 0.2222 |
| Asian/Pacific Islander; Hispanic | 0.1170 |
| Asian/Pacific Islander; Other | 0.0095 |
| Asian/Pacific Islander; White | 0.0753 |
| Black; Hispanic | 0.1062 |
| Black; Other | 0.2127 |
| Black; White | 0.1239 |
| Hispanic; Other | 0.1079 |
| Hispanic; White | 0.0291 |
| Other; White | 0.0638 |

## 7.2 – FIGURES

*Figure 2.1 Histogram of Natural Log Stop Counts by Month January 2007 – June 2016*

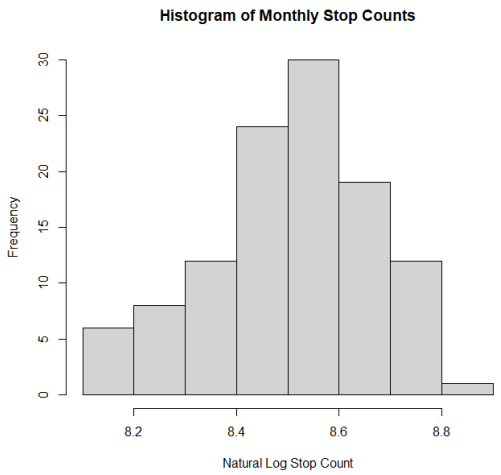



*Figure 2.2 Histogram of Natural Log Stop Counts by Month for July 2011 – June 2016*

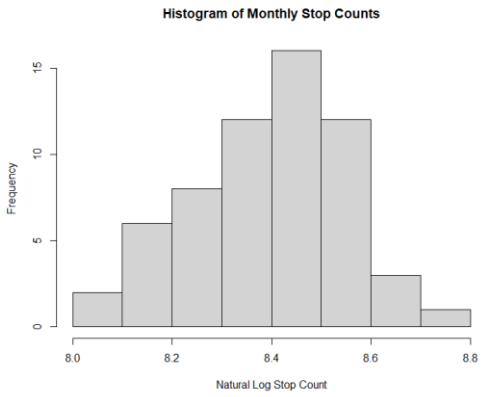



*Figure 2.3 Boxplot of Natural Log Stop Counts by Month January 2007 – June 2016*

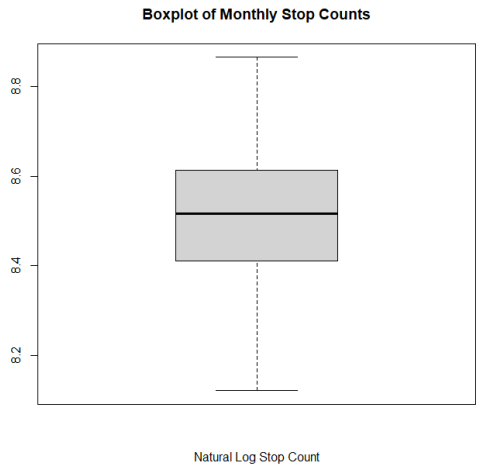


*Figure 2.4 Boxplot of Natural Log Stop Counts by Month for July 2011 – June 2016*

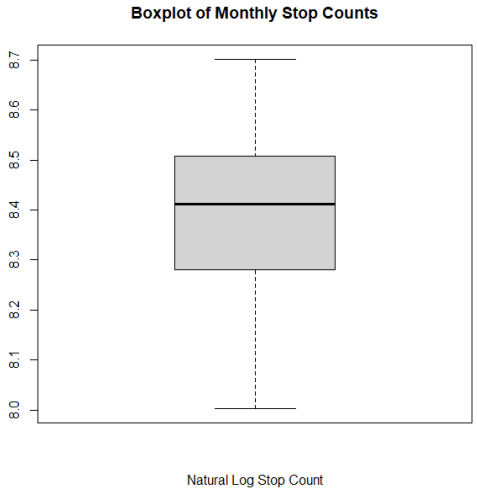



*Figure 2.5 Scatterplot of Natural Log Stop Counts by Month for July 2011 – June 2016*

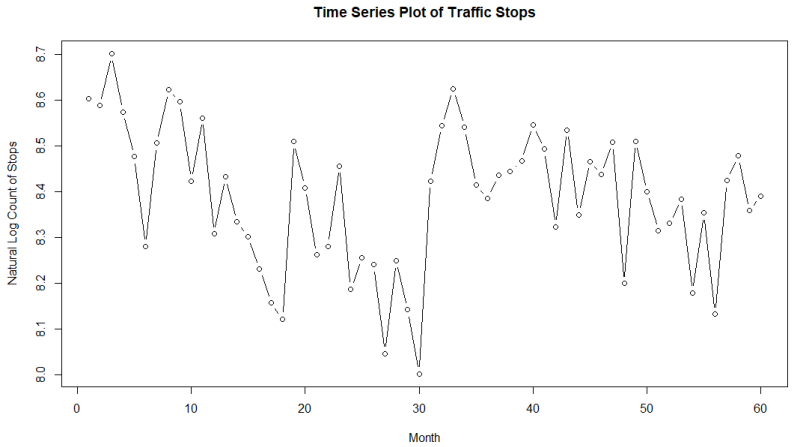



*Figure 3.1 Cook's Distance Plot*

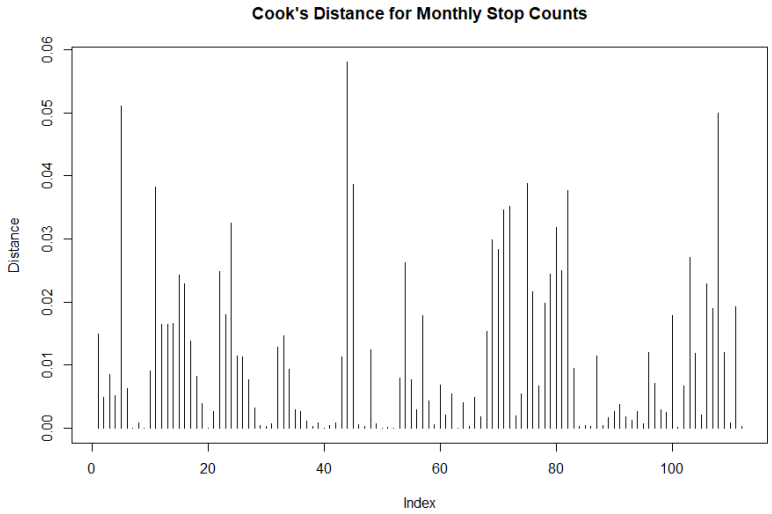


*Figure 3.2 Testing of ARIMA Model Trend, Residuals, and Normality*

There doesn't appear to be a notable trend, outliers, or change in variance over time in the standardized residual scatterplot below. The QQ plot indicates normality as the residuals remain close to a straight diagonal line. None of the lags in the ACF of residuals are significant above or below the autocorrelation threshold lines. Last, none of the residuals are significant in the Ljung-Box plot below the p-value threshold. Therefore, the ARIMA model is suitable for the data.

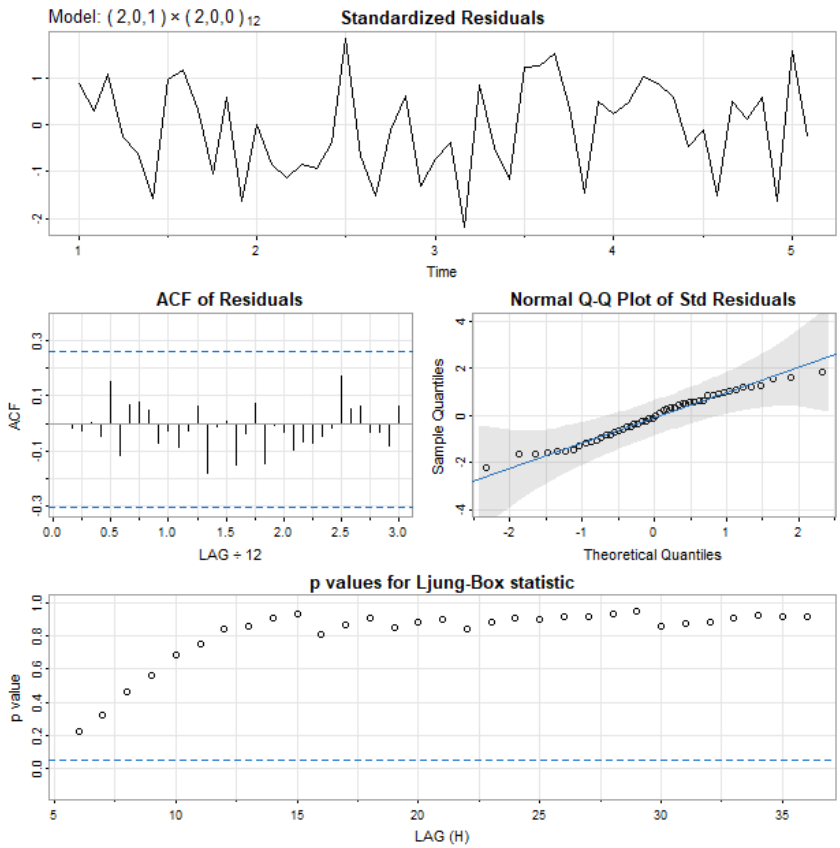
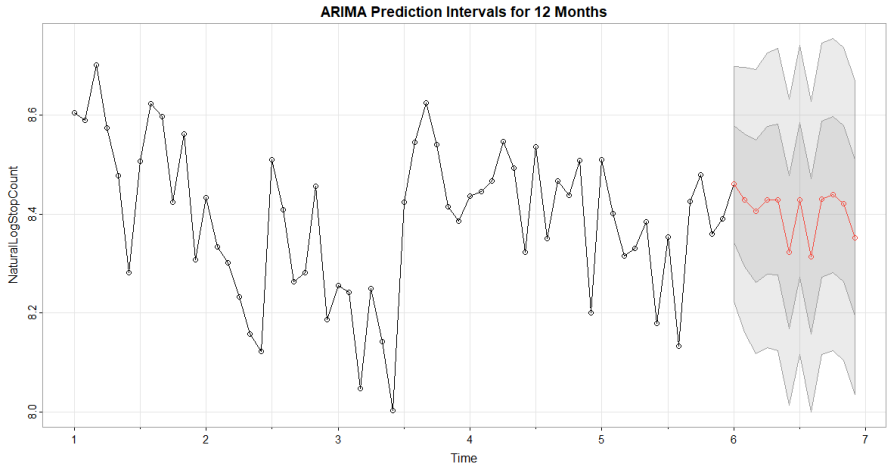



*Figure 3.5 ARIMA Prediction Intervals for 12 Months After Test Set*



7.3 – REFERENCES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